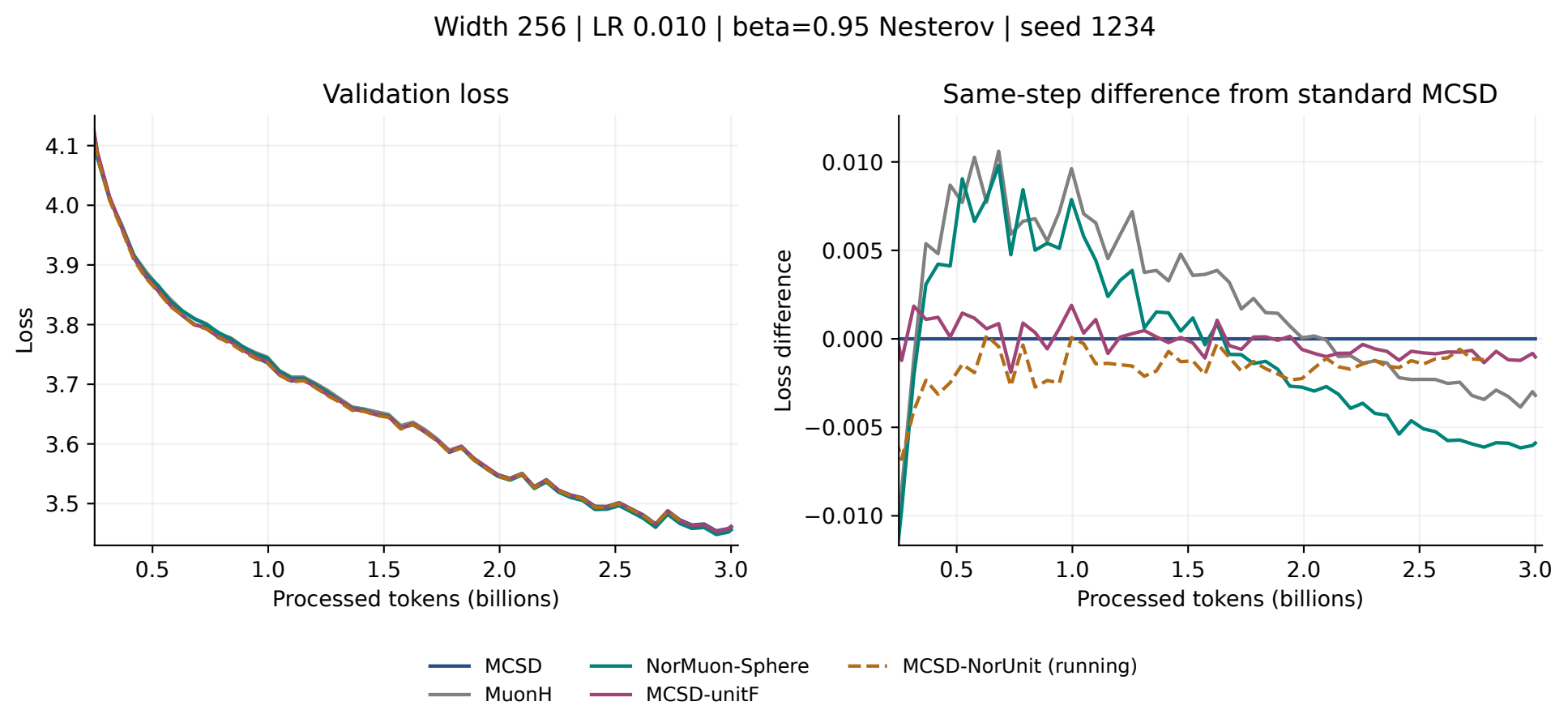
Width 256 | LR 0.010 | beta=0.95 Nesterov | seed 1234
Validation loss
Loss
4.1
4.0
3.9
3.8
3.7
3.6
3.5
0.5
1.0
1.5
2.0
2.5
3.0
Processed tokens (billions)
Same-step difference from standard MCSD
Loss difference
0.010
0.005
0.000
−0.005
−0.010
0.5
1.0
1.5
2.0
2.5
3.0
Processed tokens (billions)
MCSD
MuonH
NorMuon-Sphere
MCSD-unitF
MCSD-NorUnit (running)